

# HARSHIT GUPTA

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## PROFILE SUMMARY

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Senior Machine Learning Engineer with 6+ years of experience building production-scale AI systems, Retrieval-Augmented Generation (RAG) platforms, Agentic AI workflows, multimodal search systems, and distributed ML infrastructure. Experienced in architecting end-to-end AI products using LangGraph, AWS Bedrock, Claude, PyTorch, Spark, and cloud-native technologies across media and semiconductor domains.

## EDUCATION

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**Indraprastha Institute of Information Technology (IIIT Delhi)**

**Jul 2021 – Dec 2022**

*M.Tech in Computer Science and Engineering*

*New Delhi*

**Narula Institute of Technology**

**Aug 2014 – Jun 2018**

*B.Tech in Computer Science and Engineering*

*Kolkata, West Bengal*

## TECHNICAL SKILLS

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**Programming and ML:** Python, SQL, Machine Learning, Deep Learning, Natural Language Processing, Scikit-learn, XGBoost, TensorFlow, PyTorch, SparkML

**Generative AI and Agentic AI:** LangGraph, LangChain, Open AI, Hugging Face, Agentic AI, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Agentic RAG, Prompt Engineering, Tool Calling, Function Calling, Vector Databases, Hybrid Search

**ML Engineering and MLOps:** MLflow, LangSmith, Data Validation, Pipeline Orchestration, Model Deployment, Batch Inference, Docker, Airflow, FastAPI

**Big Data and Distributed Systems:** PySpark, HDFS, Apache Hive, Apache Sqoop, MapReduce

**Cloud Platforms:** Amazon Web Services (Bedrock, SageMaker, S3, EC2, Lambda, Athena, Comprehend, Textract), Google Cloud Platform (BigQuery, Cloud Storage, Vertex AI), Microsoft Azure (ADF, Azure SQL, Storage, Functions)

## EXPERIENCE

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**Warner Bros. Discovery**

**Hyderabad, Telangana**

*Senior Machine Learning Engineer*

*Apr 2025 – Present*

- Architected and delivered a production-grade multi-modal Document Intelligence platform using **LangGraph**, **AWS Bedrock (Claude, Titan Embeddings)**, and **PostgreSQL vector search**, automating document understanding, key-value extraction, and questionnaire completion workflows across PDFs, spreadsheets, and images, **resulting in 2,000+ annual hours of manual effort savings**.
- Led the design and deployment of scalable **unsupervised learning systems (K-Means, Hierarchical Clustering)** on 1M+ user interaction records using **PySpark and SparkML**, implementing feature engineering pipelines and model selection workflows tracked via **MLflow**.
- Architected and productionized a scene-classification platform using **AWS Textract and AWS Comprehend** for automated metadata extraction; built a serverless pipeline with **AWS Lambda to process and store outputs in AWS Athena tables**, which powered Power BI dashboards for downstream analytics and ad-targeting insights.

**Micron Technologies India**

**Hyderabad, Telangana**

*Data Science Engineer – 2 (Started as Intern)*

*Jan 2023 – Apr 2025*

- Architected and productionized a multimodal semantic search and RAG platform using **CLIP-style embeddings (PyTorch)**, **LangChain + FAISS**, and **Gemini APIs**, achieving **80% top-k cross-modal retrieval accuracy** and reducing information retrieval latency by **40%** across internal engineering workflows.
- Improved model **performance across 100+ datasets** by building automated regression and model selection pipelines using **Scikit-Learn, TensorFlow, and XGBoost**, integrating experiment tracking with **MLflow** and **hyperparameter tuning** workflows.

- Optimized large-scale SQL transformations in **GCP BigQuery**, **reducing query latency**, and engineered **parallelized preprocessing pipelines** using **PySpark**.
- Deployed model APIs via **FastAPI**, enabling scalable internal consumption across analytics teams.

## Factspan Analytics Private Limited

Bangalore, Karnataka

### *Consultant*

*Feb 2020 – Jul 2021*

- Utilized **Python, PySpark, Airflow, HDFS and Docker** to create scalable data workflows for seamless data management **improving data processing efficiency by 20%**.
- Built scalable ingestion frameworks using **Sqoop (RDBMS → HDFS)** and transformation layers in **Apache Hive**, enabling reliable data mart creation and accelerating analytics **query performance by 10%**.
- Developed **distributed SparkML** feature engineering and batch inference pipelines, eliminating costly data movement and reducing end-to-end model scoring time by **25%**.
- Orchestrated production-grade big data workflows using **Dockerized Spark jobs**, improving pipeline reliability and reducing operational failures by **10%**.

## Capgemini

Bangalore, Karnataka

### *Senior Analyst (Started as Analyst)*

*Aug 2018 – Feb 2020*

- Streamlined data processing with end-to-end **Azure Data Factory pipelines integrating Azure SQL, Storage, and Azure Functions**, enabling scalable ETL workflows across enterprise data systems, reducing processing time by **15%**.
- Streamlined data validation and transformation processes using **Python and PowerShell**, improving reliability of downstream ML use cases, enhancing workflow efficiency by **25%**.
- Managed Sharded storage and retrieval systems in **HDFS and SQL Server**, improving storage and retrieval efficiency by **30%**.

## ACCOMPLISHMENTS

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- GATE 2021 (CSE): AIR **1432 (Top 1.3%)** among 101,922 candidates; 4 times GATE qualifier with strong CS fundamentals.
- Solved **800+** algorithmic questions on Leetcode, reflecting strong algorithmic thinking and rigor.

## AWARDS

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- Culture Champion Award - Micron Technology
- Bravo Award – Micron Technology - Recognized for exceptional performance and successful delivery of a complex data science / AI project.